

Ben Sanati

PhD Student, University of Edinburgh

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Research Interests. My interests lie in the theoretical foundations of machine learning, with a focus on developing principled approaches to learning under uncertainty. My work explores continual learning, bounded rationality, epistemology, and subjective perspectives on probability theory.

Education

- 2024–2028 **Ph.D in Machine Learning Systems**, UNIVERSITY OF EDINBURGH.
Advisors: David Abel, Amos Storkey.
- 2019–2023 **MEng Electronic Engineering with AI**, UNIVERSITY OF SOUTHAMPTON.
First Class Honours
Advisors: Jonathon Hare, Geoff V. Merrett
Thesis: *Improving Inference Efficiency with Dynamic DNNs*
Thesis: *Detecting, Mapping, and Verifying Signage*

Professional Experience

- 03/24–07/24 **Senior Research Assistant**, [project page](#), UNIVERSITY OF SOUTHAMPTON.
◦ Sole research associate on a **£25k Innovate UK-funded** industry collaboration.
◦ Designed and implemented a mobile 3D SLAM system and end-to-end data lake architecture.
◦ Delivered technical findings and research outcomes to industry stakeholders.
- 07/23–09/23 **Data Science Intern**, [hackathon](#), CIRIUM, RELX GROUP.
◦ Awarded a data science internship following **success in a competitive ML hackathon**.
◦ Delivered spatio-temporal ML models for large-scale aviation forecasting tasks.
- 06/22–09/22 **Undergraduate Research Scholar**, [project page](#), UNIVERSITY OF SOUTHAMPTON.
◦ Awarded a **£5k research scholarship** for academic excellence by a faculty scholarship panel.
◦ Investigated computationally efficient computer-vision methods for object detection.
◦ Presented research findings to students and faculty members.

Awards

- 2024 EPSRC Machine Learning Systems Scholarship
2022 Undergraduate Research Scholarship

Publications

Forgetting is Everywhere

Ben Sanati, Thomas L. Lee, Trevor McInroe, Aidan Scannell, Nikolay Malkin, David Abel, Amos Storkey.
[In Submission.](#)

Talks, Reviewing, and Outreach

- Talks 2026: Foundations of Reinforcement Learning. [Reading Group, Imperial College London.](#)
Forgetting is Everywhere. [IML Workshop, University of Edinburgh.](#)
2025: The fundamental problem of forgetting. [ANC Workshop, University of Edinburgh.](#)
- Reviewing 2026: Conference on Lifelong Learning Agents – [CoLLAs](#).
Continual Adaptation at Scale – [CATS@ICML](#).
2025: Finding the Frame – [RLC](#).

Further Development 2026: Reinforcement Learning Summer School, *ELLIS Unit Milan, Politecnico di Milano*.
Mathematical Foundations of Probabilistic AI. *ProbAI Hub Winter School, University of Warwick*.